

1) Launch one AWS EC2 and install Jenkins

Objective:

To launch an AWS EC2 instance and install Jenkins on it for CI/CD automation. Jenkins installation on AWS EC2 typically requires Java and port 8080 to be open in the security group.

Commands Executed:

```
sudo apt update
```

```
sudo apt install openjdk-21-jre -y
```

```
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee  
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
```

```
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-  
stable binary/ | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

```
sudo apt-get install jenkins -y
```

```
sudo systemctl start jenkins
```

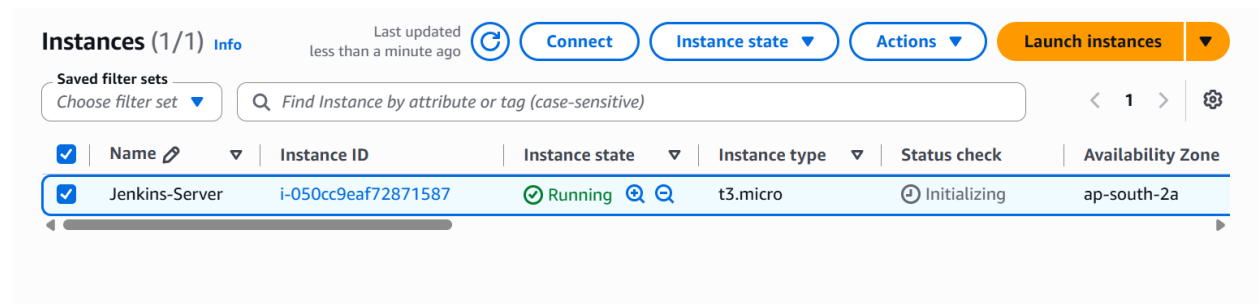
```
sudo systemctl status jenkins
```

Command Output:

- Java installed successfully.
- Jenkins package installed successfully.
- Jenkins service started successfully.
- systemctl status showed Jenkins as **active (running)**.

Evidence:

Step 1 : Lanuch a ec2 Instances in Aws



The screenshot displays the AWS Management Console's 'Instances' page. At the top, it shows 'Instances (1/1)' with an 'Info' link and a refresh button. Below this, there are buttons for 'Connect', 'Instance state', 'Actions', and a prominent orange 'Launch instances' button. A search bar is present with the placeholder text 'Find Instance by attribute or tag (case-sensitive)'. The main table lists the instance 'Jenkins-Server' with ID 'i-050cc9eaf72871587', which is in a 'Running' state (indicated by a green checkmark). The instance type is 't3.micro' and it is located in the 'ap-south-2a' availability zone. The status check shows 'Initializing'.

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☒	Jenkins-Server	i-050cc9eaf72871587	Running	t3.micro	Initializing	ap-south-2a

Step 2: type `sudo apt update`.

```
ubuntu@ip-172-31-14-58:~$ sudo apt update
Hit:1 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [248 kB]
Get:7 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [64.2 kB]
```

Step 3: type `sudo apt install openjdk-17-jre -y` to install Java.

```
ubuntu@ip-172-31-14-58:~$ sudo apt install fontconfig openjdk-17-jre -y
Installing:
  fontconfig  openjdk-17-jre

Installing dependencies:
  adwaita-icon-theme      libasound2-data      libglib2.0-0      libpixman-1-0      libxi6
  alsa-topology-conf      libasound2t64        libglvnd0         librsvg2-2         libxinerama1
  alsa-ucm-conf           libatk-bridge2.0-0t64 libglx-mesa0      librsvg2-common    libxkbfile1
  at-spi2-common          libatk-wrapper-java  libglx0           libsharpyuv0       libxmu6
  at-spi2-core            libatk-wrapper-java-jni libglibc2-0       libsm6             libxpm4
  bubblewrap              libatk1.0-0t64       libgraphite2-3    libthai-data       libxrandr2
```

Step 4: type `java -version` to see java version

```
ubuntu@ip-172-31-14-58:~$ java -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-14-58:~$
```

Step 5 : type `sudo apt-get install jenkins -y` to install and `Jenkins -version` to see Jenkins version.

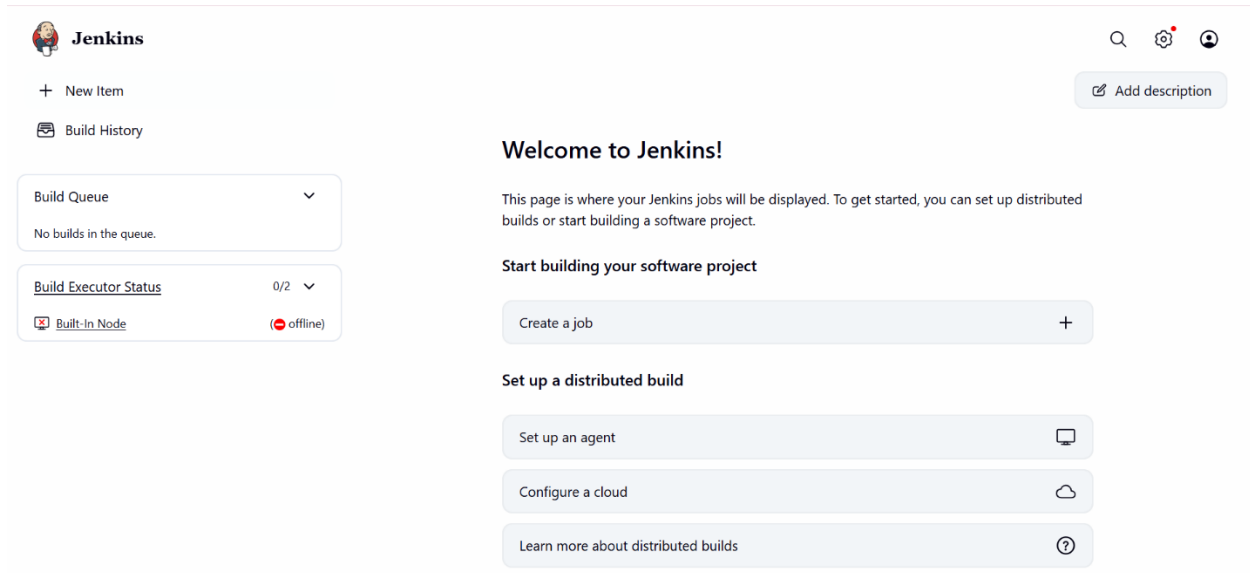
```
ubuntu@ip-172-31-14-58:~$ jenkins --version
2.555.3
ubuntu@ip-172-31-14-58:~$
```

Step 6 : Enter `sudo systemctl start Jenkins` to start and `sudo systemctl status Jenkins` to see Jenkins Running Are Not.

```
ubuntu@ip-172-31-14-58:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-29 08:01:58 UTC; 14s ago
     Invocation: a69530775efe433da03920b62f1252f4
   Main PID: 8086 (java)
    Tasks: 41 (limit: 627)
  Memory: 320.6M (peak: 360.4M)
     CPU: 20.512s
   CGroup: /system.slice/jenkins.service
           └─8086 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]>
Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]> *****
Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]> *****
Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]> *****
Jun 29 08:01:58 ip-172-31-14-58 jenkins[8086]: 2026-06-29 08:01:58.137+0000 [id=34] INFO jenkins.InitReactorRunner$1#onAttain
```

Step 7: Enter the public Ip And enter default process.



Key Learnings:

- Jenkins requires Java before installation.
- Port 8080 must be opened in AWS security group.
- Jenkins on EC2 is commonly installed using the official APT repository.

2) Setup Apache Tomcat on Ubuntu Virtual Machine

Objective:

To install Apache Tomcat on an Ubuntu virtual machine for running Java web applications. Tomcat needs Java and is usually tested on port 8080.

Step 1: Update Packages

Commands Executed:

```
sudo apt update
```

Step 2: Install Java

```
sudo apt install open-jdk-21-jdk -y
```

java -version

Step 3: Download Apache tomcat

Wget <https://tomcat.apache.org/download-10.cgi>

Step 4: Extract Tomcat

tar -xvzf apache-tomcat-10.1.xx.tar.gz

Step 5: Move Tomcat

cd /opt/tomcat/bin

Step 6 : Start Tomcat

Cd /opt/tomcat/bin

chmod +x *.sh

./startup.sh

Step 7 : Verify

sudo systemctl status tomcat

open browser:

http://8080

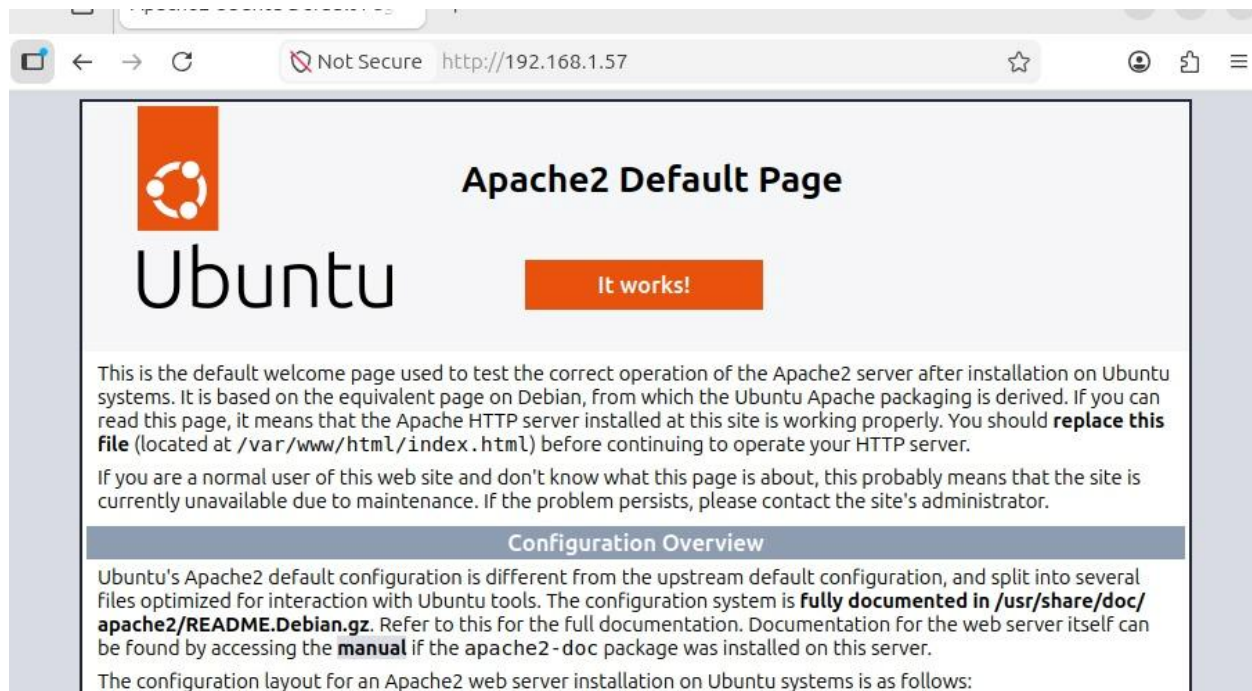
Command Output:

- Ubuntu packages updated successfully.
- JDK installed successfully.
- Tomcat started successfully.

Evidence:

```
root@ubuntu2604:/opt/apache-tomcat-9.0.119/bin# systemctl status tomcat
● tomcat.service - Apache Tomcat 9
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-06-27 08:08:00 EDT; 2min 22s ago
  Invocation: f42acb23ead44c4abce458d8ca6bc185
    Process: 21508 ExecStart=/opt/apache-tomcat-9.0.119/bin/startup.sh (code=exited, status=0/SUCCESS)
   Main PID: 21515 (java)
      Tasks: 37 (limit: 4021)
     Memory: 120.4M (peak: 120.8M)
        CPU: 5.941s
    CGroup: /system.slice/tomcat.service
            └─21515 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.util.logging.config.file=/opt/apache-tomcat-9.0.119/conf/logging.pr

Jun 27 08:08:00 ubuntu2604 systemd[1]: Starting tomcat.service - Apache Tomcat 9...
Jun 27 08:08:00 ubuntu2604 startup.sh[21508]: Tomcat started.
Jun 27 08:08:00 ubuntu2604 systemd[1]: Started tomcat.service - Apache Tomcat 9.
lines 1-15/15 (END)
```



Key Learnings:

- Tomcat depends on Java.
- Correct file ownership is important.
- Port 8080 is used to verify the server in the browser.

3) Setup Docker Desktop on local Windows

Objective:

To install Docker Desktop on a Windows system for running containers locally. Docker Desktop on Windows often uses WSL 2 and may require a restart after installation.

Commands Executed:

```
wsl --install
```

```
wsl --update
```

Download Docker Desktop And Install.

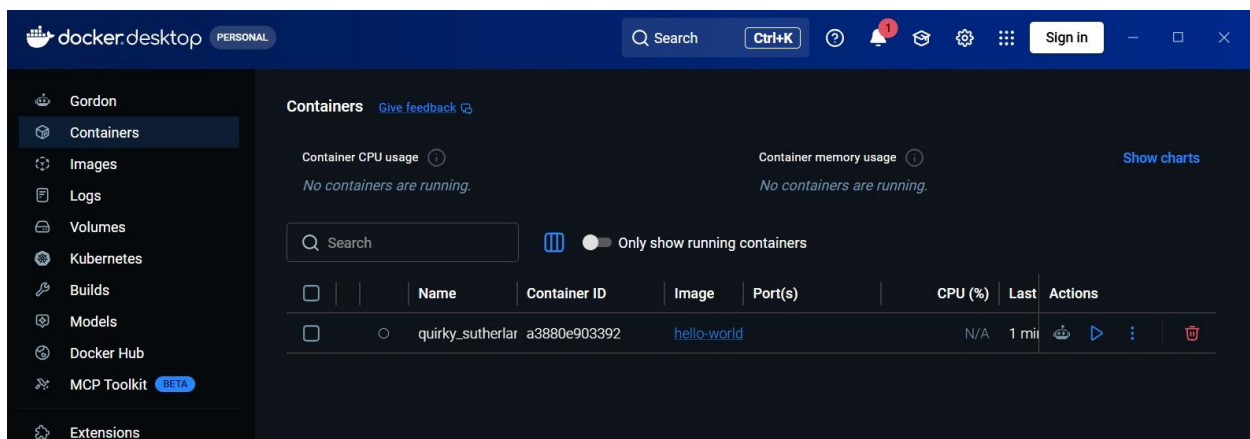
Command Output:

- wsl --install installs WSL components if needed.

- wsl --update updates the WSL kernel.
- Docker Desktop installation completes successfully.

Evidence:

```
Create a default Unix user account: srihari
Setup Docker Desktop on your local windows.
windows.ng: Windows Subsystem for Linux 2.7.10
Installing: Windows Subsystem for Linux 2.7.10
Windows Subsystem for Linux 2.7.10 has been installed.
The operation completed successfully.
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Setup Docker Desktop on your local windows. Create a default Unix user account: srihari
```



Key Learnings:

- WSL 2 support is important for Docker Desktop.
- Restart may be required after installation.
- docker run hello-world is the easiest verification step.